The Induction Principle — problem 8.7

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We build an exponential tower

$$\sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdot^{\cdot}}}}$$

by defining $a_0 = 1$, and $a_{n+1} = \sqrt{2}^{a_n}$, $(n \in \mathbb{N}_0)$. Show that the sequence a_n is monotonically increasing and bounded above by 2.